

Xuanzhuo Liu

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RESEARCH INTERESTS

I am interested in vision-language-action foundation models that connect multimodal reasoning with reliable action generation for autonomous systems. My focus is on reinforcement learning post-training, reward and value learning, and constrained improvement of pretrained generative policies.

EDUCATION

Stanford University

M.S. Student in Electrical Engineering · Advisor: Chelsea Finn

Expected Jun. 2028

Stanford, CA

The Chinese University of Hong Kong, Shenzhen

B.S. in Data Science with Honors, School of Data Science – Nobel Class

GPA: 3.94/4.00 · Rank: 1/315 · Advisor: Prof. Tongxin Li, Prof. Zizhuo Wang

Jun. 2026

Shenzhen, China

University of Oxford

Visiting Student in Mathematics and Computer Science, St Hilda's College

GPA: 3.97/4.00 · Earned First-Class marks in six graduate-level courses

Oct. 2024 – Jun. 2025

Oxford, UK

RESEARCH EXPERIENCE

AgiBot Robotics – Embodied Research Center

Algorithm Developer & Research Intern

May 2026 – Sept. 2026

Shanghai, China

• RetraQ: Manifold-Constrained Critic Guidance for Flow Policies in RL

[\[Project Website\]](#)

- **Geometry-Aware Generative Modeling:** Developed critic guidance for pretrained flow-matching policies, using the denoising Jacobian to filter update directions and denoising-based retraction to keep refined actions near the data-supported manifold.
- **Controlled Data Collection:** Designed forked rollouts at critical manipulation states to capture successful and failed trajectories; trained action-conditioned critics to guide policy refinement.
- **Real-Robot Evaluation:** Improved $\pi_{0.5}$ phone-charging success from 55.0% to **76.7%** with test-time guidance, reaching **98.3%** success within 30 minutes of online training.

• General Reward Model

[\[One-Shot OOD Demo\]](#)

- **Visual and Temporal Understanding:** Developed a reference-conditioned vision-language reward model for subtask completion, temporal progress, and task success, enabling one-shot generalization to unseen tasks from a single demonstration.
- **Real-World Data Pipelines:** Contributed to data pipelines supporting SFT on 3,000+ hours of data, including 30,000+ egocentric and UMI demonstration episodes.
- **Reward-Guided Learning:** Trained MC and IQL critics with learned dense rewards to capture failure and recovery signals; integrated the critics into AWR for real-robot manipulation.
- **Simulation and Real-Robot Results:** Project-level evaluations improved success from 30% to **70%** on LIBERO-10 Task 8 with online DSRL, and from 15% to **75%** on real-robot phone charging with offline VINE RL.

Microsoft Research Asia – Machine Learning Group

Research Intern – Stars of Tomorrow Program [\[Blog Highlight\]](#)

Jul. 2025 – Feb. 2026

Beijing, China

- Built and pretrained a **4B-parameter vision-language-action model** on over **10,000 hours** of manipulation data for multimodal perception and action generation.
- Introduced minimum-jerk regularization to improve temporal smoothness of generated actions, reducing action variance by 25% for more stable robot control.
- Adapted the pretrained model to 10 tasks with 150 episodes per task, achieving 80% average success and demonstrating data-efficient transfer.
- Achieved top-two globally in the 2025 IROS Manipulation Challenge and received the Rising Tech Talent award.

Oxford Robotics Institute

Oct. 2024 – Jul. 2025

Research Assistant · Advisor: Prof. Shimon Whiteson

Oxford, UK

- Proposed a dual-manifold projection scheme for model-predictive control that separates transition dynamics from feasibility constraints, reducing storage requirements by 20% while maintaining real-time control at 30 Hz.
- Implemented Differential Transformer layers to suppress attention noise and emphasize task-relevant signals, achieving over 70% success on SimplerEnv simulation tasks.
- Developed a PDE-based framework to analyze information flow in Transformer policies and identify architectural bottlenecks.

Shenzhen Research Institute of Big Data

Sept. 2023 – Aug. 2024

Research Assistant · Advisor: Prof. Haizhou Li

Shenzhen, China

- Designed controlled experiments and statistical pipelines for visual-auditory perception; analyzed cross-modal attention using eye tracking and presented the work at ICSR 2024.
- Used large language models to extract structured diagnostic rules from clinical text and medical guidelines.
- Developed a sparse Transformer with two-stage reinforcement learning for clinical perception and reasoning, achieving approximately 67% diagnostic-decision accuracy on MIMIC.

PUBLICATIONS & MANUSCRIPTS

UNDER REVIEW RetraQ: Manifold-Constrained Critic Guidance for Flow Policies in RL.

Xuanzhuo Liu, Chen Wu, Shuai Yuan, Chuankang Li, Maoqing Yao, and Guanghui Ren. [Project Website]

PEER-REVIEWED The Impact of Synchronized Visual and Auditory Attention on Human Perception.

Lichuan Jiang*, Jiani Zhong*, **Xuanzhuo Liu**[†], Siqu Cai, and Haizhou Li.

International Conference on Social Robotics (ICSR 2024), pp. 41–50, Springer, 2025. **Best Student Paper Award.** [Paper]

TECH REPORT Reference-Conditioned Reward Modeling for Generalizable Robot Learning.

Hecheng Wang, **Xuanzhuo Liu**, Xinchun Li, Yexin Li, Chuankang Li, Maoqing Yao, and Guanghui Ren.

HONORS & AWARDS

School of Data Science Outstanding Graduate Award	2026
Presidential Award for Outstanding Students , highest honor for university graduates	2026
Microsoft Stars of Tomorrow , Microsoft Research Asia	2025
Rising Tech Talent , IROS AgiBot Manipulation Competition	2025
Best Undergraduate Teaching Fellow , CUHK(SZ)	2025
Undergraduate Research Award , CUHK(SZ)	2024
Best Student Paper , International Conference on Social Robotics (ICSR)	2024
Class A Academic Scholarship , Top 1% in the School of Data Science	2024
Dean's List , School of Data Science	2023-2025
Diligentia Full-tuition Scholarship & Special Grant	2022-2025

TEACHING

Undergraduate Teaching Assistant, CUHK-Shenzhen

DDA3020, Machine Learning Fall 2025

MAT2041, Linear Algebra and Its Applications Fall 2024

STA4001, Stochastic Process Spring 2024

CSC3100, Data Structure and Algorithms Fall 2023

TECHNICAL SKILLS

Programming & ML: Python, PyTorch, JAX, C++, distributed training

Simulation & Robotics: Isaac Gym/Lab, MuJoCo, ROS/ROS2, simulation evaluation, real-robot deployment

Research Methods: flow-matching policies, geometry-aware guidance, multimodal data pipelines, reinforcement learning

Languages: English (TOEFL iBT 110, IELTS 8), Chinese